# Incident investigation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

## One complete incident evidence and hypothesis record.

Keep permitted timeline, versions, containment, candidate explanation, test conditions and unresolved evidence together.

**Keep one habit:** Recall → W1–W3/W5 → independent permission incident → investigation project → delayed return.

MAIN LESSON · SECTION 1

## Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Monitoring quality drift. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Incident investigation preserves a timeline and distinguishes **observation** from hypothesis. Repair before **evidence** collection can erase clues.

- Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate **containment** from diagnosis. Compare candidate explanations against **evidence**.

### Monitoring quality drift

> **My remembered idea, example and correction**
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### Deployment environments

> **My remembered idea, example and correction**
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### Data retention and deletion

> **My remembered idea, example and correction**
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### Lesson reference for workbook practice

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MAIN LESSON · SECTION 5

# W1 · Explain the current example

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Incident investigation records observed times, versions, requests, authorized actions and outcomes using minimized permitted evidence. Separate observations from hypotheses, and immediate containment from diagnosis. Main supplied timeline: corpus c2 publishes at 09:00; response uses stale c1 at 09:05; cache disabled at 09:06.

  Trusted declared weight version remains w7, contradicting the reported weight-version change under that metadata assumption. Stale cache is a candidate, not a proven exact cause. A controlled local deliberately buggy query-only cache gives c1 when enabled and c2 when bypassed; other fixtures stay fixed. This is a possible mechanism demonstration, not a live reproduction.

  Separate picture: 14:00 version 6 room D; 14:05 version 5 room C used; 14:06 mismatch report. Changed trace retrieves version 6 but drafts room C, so inspect handoff, answer construction and checks instead of assuming old-source retrieval. Missing events may change conclusions; temporal order alone proves no cause. Contain promptly while authorized minimized evidence is captured where feasible.

  No private payloads, real logs, live model runs or service changes occur in the code.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**My reasoning and evidence**

MAIN LESSON · SECTION 6

## W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Incident investigation records observed times, versions, requests, authorized actions and outcomes using minimized permitted evidence. Separate observations from hypotheses, and immediate containment from diagnosis. Main supplied timeline: corpus c2 publishes at 09:00; response uses stale c1 at 09:05; cache disabled at 09:06.

  Trusted declared weight version remains w7, contradicting the reported weight-version change under that metadata assumption. Stale cache is a candidate, not a proven exact cause. A controlled local deliberately buggy query-only cache gives c1 when enabled and c2 when bypassed; other fixtures stay fixed. This is a possible mechanism demonstration, not a live reproduction.

  Separate picture: 14:00 version 6 room D; 14:05 version 5 room C used; 14:06 mismatch report. Changed trace retrieves version 6 but drafts room C, so inspect handoff, answer construction and checks instead of assuming old-source retrieval. Missing events may change conclusions; temporal order alone proves no cause. Contain promptly while authorized minimized evidence is captured where feasible.

  No private payloads, real logs, live model runs or service changes occur in the code.

- At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. Show the intermediate steps and give the conclusion with its scope.

### My reasoning and evidence

# W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

## Complete task and evidence

- Incident investigation records observed times, versions, requests, authorized actions and outcomes using minimized permitted evidence. Separate observations from hypotheses, and immediate containment from diagnosis. Main supplied timeline: corpus c2 publishes at 09:00; response uses stale c1 at 09:05; cache disabled at 09:06.

  Trusted declared weight version remains w7, contradicting the reported weight-version change under that metadata assumption. Stale cache is a candidate, not a proven exact cause. A controlled local deliberately buggy query-only cache gives c1 when enabled and c2 when bypassed; other fixtures stay fixed. This is a possible mechanism demonstration, not a live reproduction.

  Separate picture: 14:00 version 6 room D; 14:05 version 5 room C used; 14:06 mismatch report. Changed trace retrieves version 6 but drafts room C, so inspect handoff, answer construction and checks instead of assuming old-source retrieval. Missing events may change conclusions; temporal order alone proves no cause. Contain promptly while authorized minimized evidence is captured where feasible.

  No private payloads, real logs, live model runs or service changes occur in the code.

- A report omits this qualification: A plausible narrative needs controlled or record-based confirmation and must preserve unknowns.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

# W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

## Complete task and evidence

- Incident investigation records observed times, versions, requests, authorized actions and outcomes using minimized permitted evidence. Separate observations from hypotheses, and immediate containment from diagnosis. Main supplied timeline: corpus c2 publishes at 09:00; response uses stale c1 at 09:05; cache disabled at 09:06.

  Trusted declared weight version remains w7, contradicting the reported weight-version change under that metadata assumption. Stale cache is a candidate, not a proven exact cause. A controlled local deliberately buggy query-only cache gives c1 when enabled and c2 when bypassed; other fixtures stay fixed. This is a possible mechanism demonstration, not a live reproduction.

  Separate picture: 14:00 version 6 room D; 14:05 version 5 room C used; 14:06 mismatch report. Changed trace retrieves version 6 but drafts room C, so inspect handoff, answer construction and checks instead of assuming old-source retrieval. Missing events may change conclusions; temporal order alone proves no cause. Contain promptly while authorized minimized evidence is captured where feasible.

  No private payloads, real logs, live model runs or service changes occur in the code.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

MAIN LESSON · SECTION 4

# Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- 10:00 permissions config changes; 10:03 restricted read; model version unchanged. Name a plausible cause and one test.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Independent case: Apply a new case independently

**My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

## Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

### Complete task and evidence

- Add a labelled incident investigation **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of incident investigation.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

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MAIN LESSON · SECTION 11

# Feedback · Repair and return

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Use W2: Trace the worked case for Incident investigation. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

## Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

### Complete task and evidence

- Use the worked case for Incident investigation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

## Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

### Complete task and evidence

- 10:00 permissions config changes; 10:03 restricted read; model version unchanged. Name a plausible cause and one test.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

## Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- An **incident timeline**. After a schedule update, a service gives an old classroom location.
- 14:00 version 6 publishes room D; 14:05 answer uses version 5 room C; 14:06 user reports mismatch.
- Order the three incident records and locate the old **evidence** version. How does the investigation change if the trace instead retrieves version 6 but drafts room C?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.




Follow the labelled picture with the matching explanation. Open the vector diagram

**My picture-based explanation and changed-case prediction**

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

# Reference · Project milestone

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |

| Evidence to keep | My record |
| --- | --- |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

# Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

## First supported question and optional hint

- An **incident timeline**. After a schedule update, a service gives an old classroom location.
- Which labelled step matches this detail? Reproduce version selection with controlled fixtures before naming the exact bug.
- 1. Test explanation
- 2. Order events
- 3. Locate stale **evidence**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Test explanation Reproduce version selection with controlled fixtures before naming the exact bug. A timeline separates observed events from a proposed cause.

**Evidence** can locate a mismatch without yet explaining why it happened.

## Changed-case question and optional hint

- Changed-case practice: Trace instead shows version 6 retrieved and room C drafted. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Investigate drafting or validation before blaming caching.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Investigate drafting or validation before blaming caching. The mismatch now appears after retrieval.

## Repair and retain the record

Review Monitoring quality drift (ai-6/25). Goal: Explain the **mechanism** and terminology of monitoring quality drift. Short practice: Compute both clear and ambiguous slice rates for A and B.

Which slice weakens, and how should that slice be reported when no ambiguous questions are observed?. Return to this week and explain what you changed.

Review Deployment environments (ai-6/23). Goal: Explain the **mechanism** and terminology of deployment environments. Short practice: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?. Return to this week and explain what you changed.

Review Data retention and deletion (ai-6/20). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Apply the seven-day boundary to T1 and T2.

How many T2 locations need verification, and may deletion be called complete if its backup remains?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

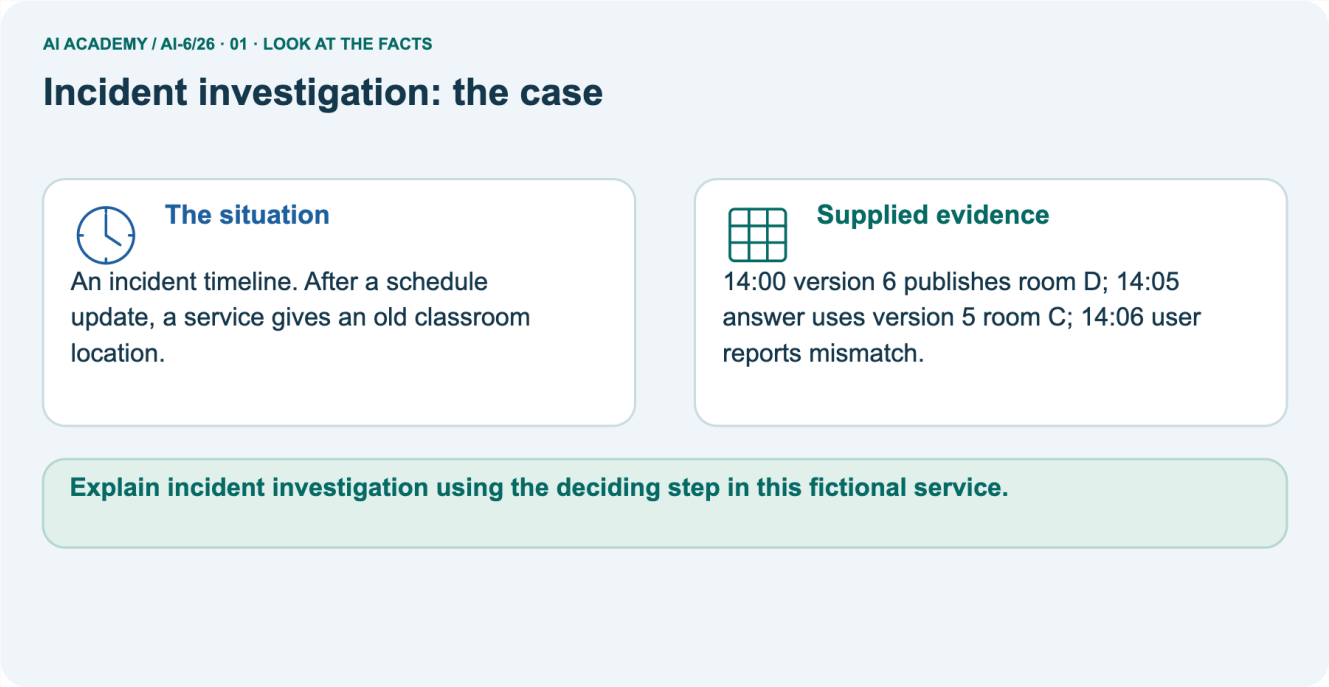
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

## Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

## 01. Incident investigation: the case



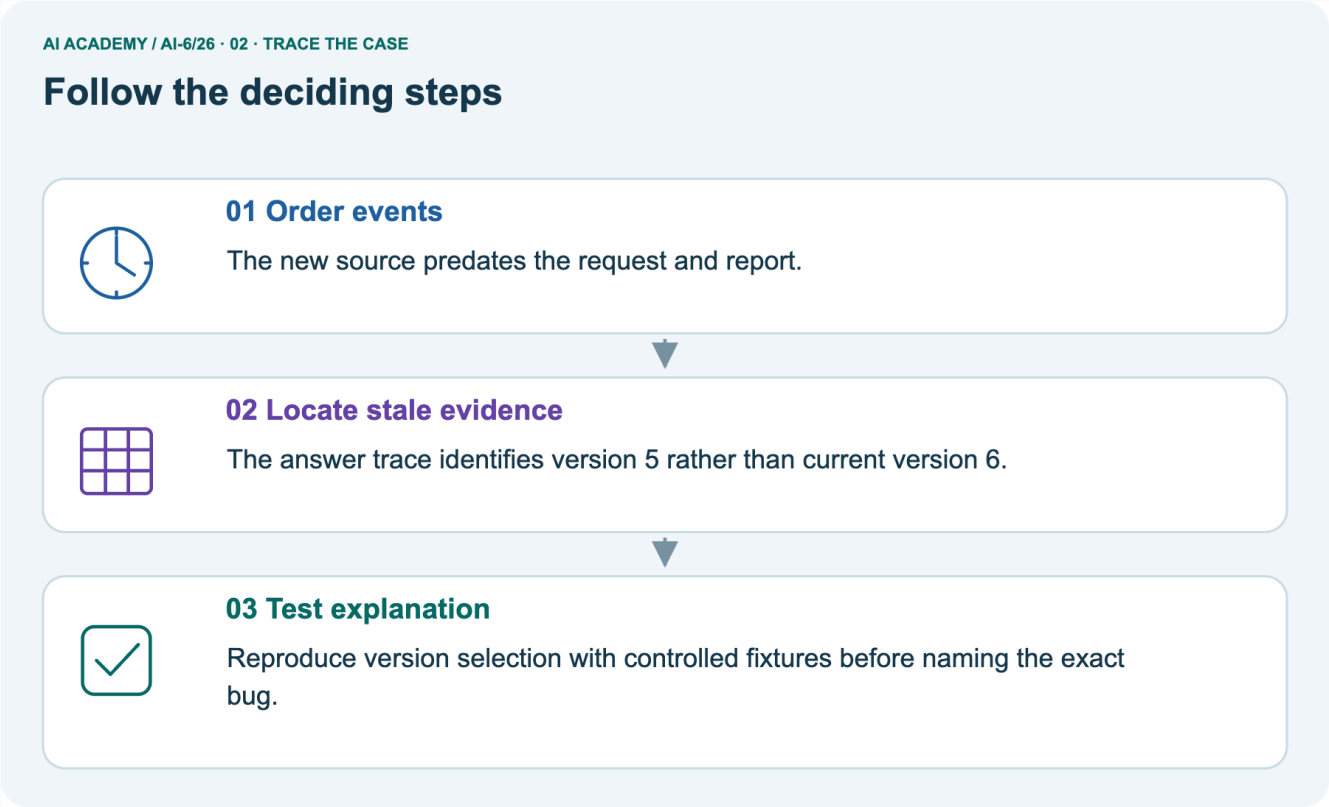

Follow the labelled picture with the matching explanation. Open the vector diagram

An **incident timeline**. After a schedule update, a service gives an old classroom location. Supplied facts: 14:00 version 6 publishes room D; 14:05 answer uses version 5 room C; 14:06 user reports mismatch.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



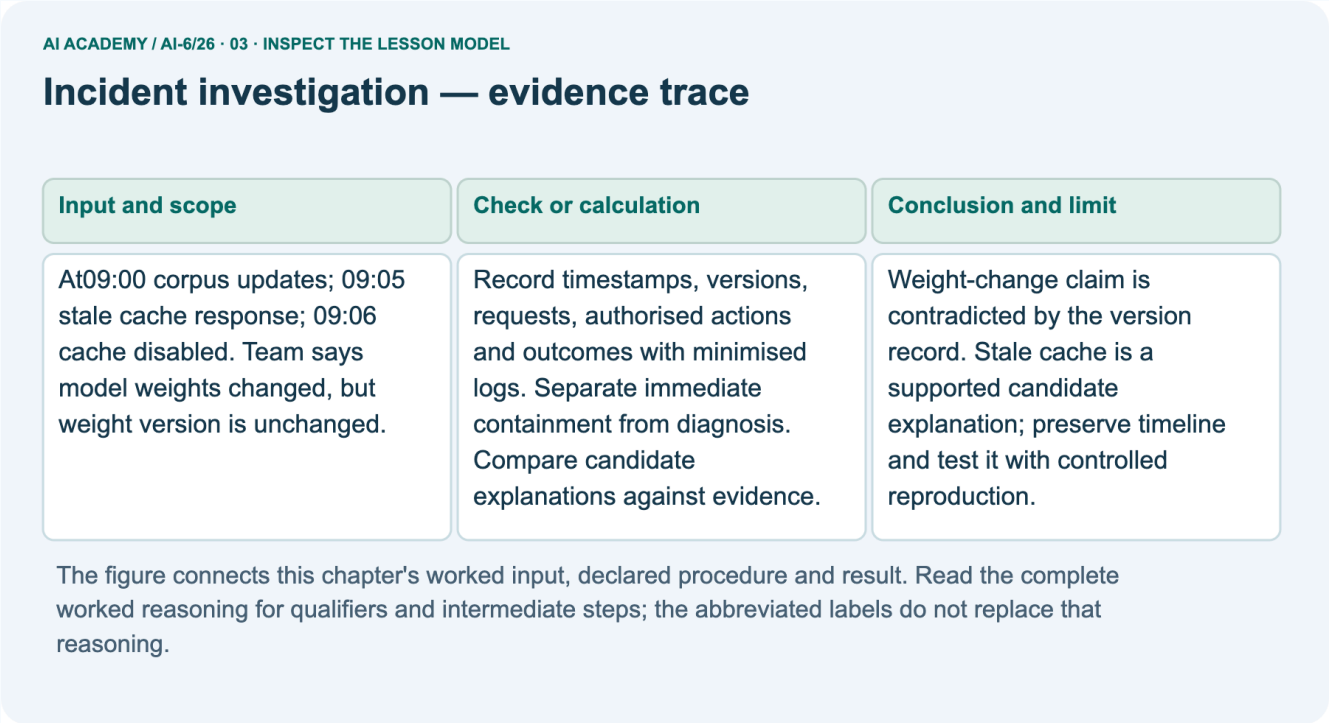

Follow the labelled picture with the matching explanation. Open the vector diagram

Order events: The new source predates the request and report. → Locate stale **evidence**: The answer trace identifies version 5 rather than current version 6. → Test explanation: Reproduce version selection with controlled fixtures before naming the exact bug.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Incident investigation — evidence trace



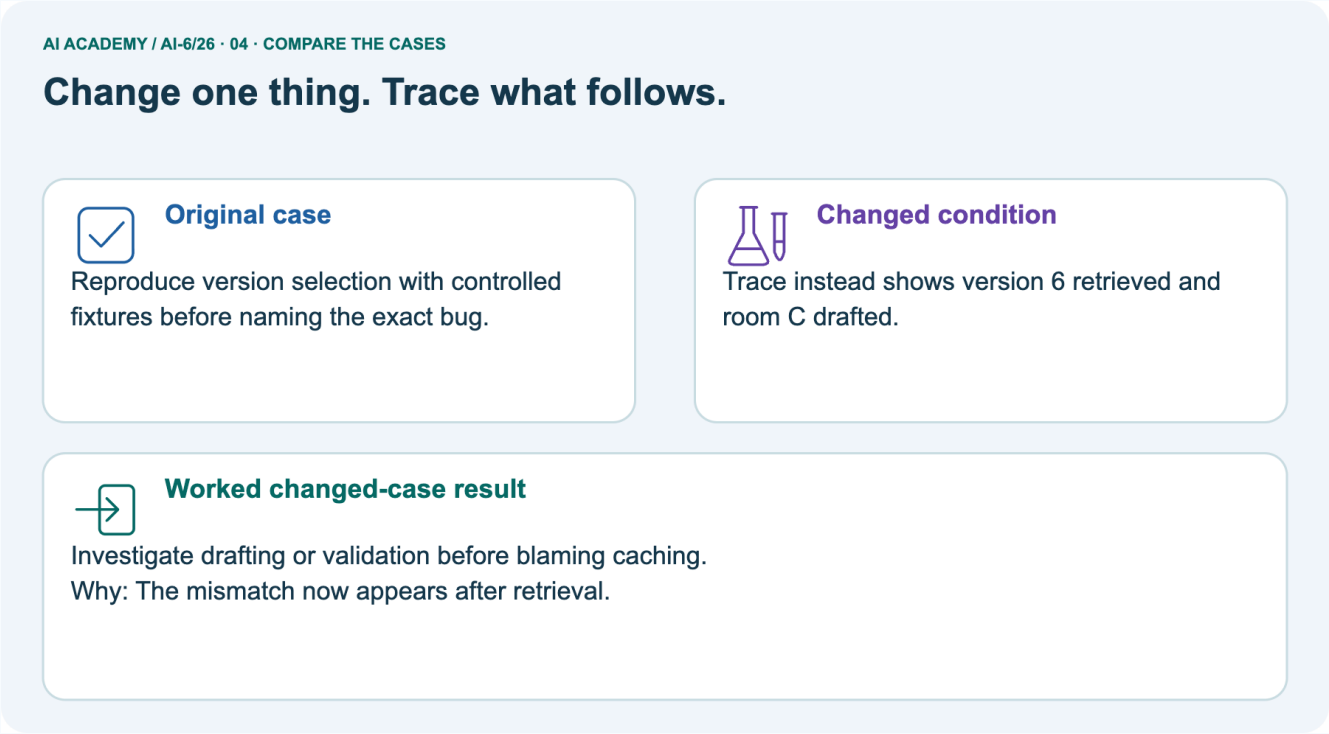

Follow the labelled picture with the matching explanation. Open the vector diagram

Incident investigation — **evidence** trace. At 09:00 corpus updates; 09:05 stale cache response; 09:06 cache disabled. Team says model weights changed, but weight version is unchanged. | Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate **containment** from diagnosis. Compare candidate explanations against **evidence**. | Weight-change claim is contradicted by the version record. Stale cache is a supported candidate explanation; preserve timeline and test it with controlled reproduction.

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

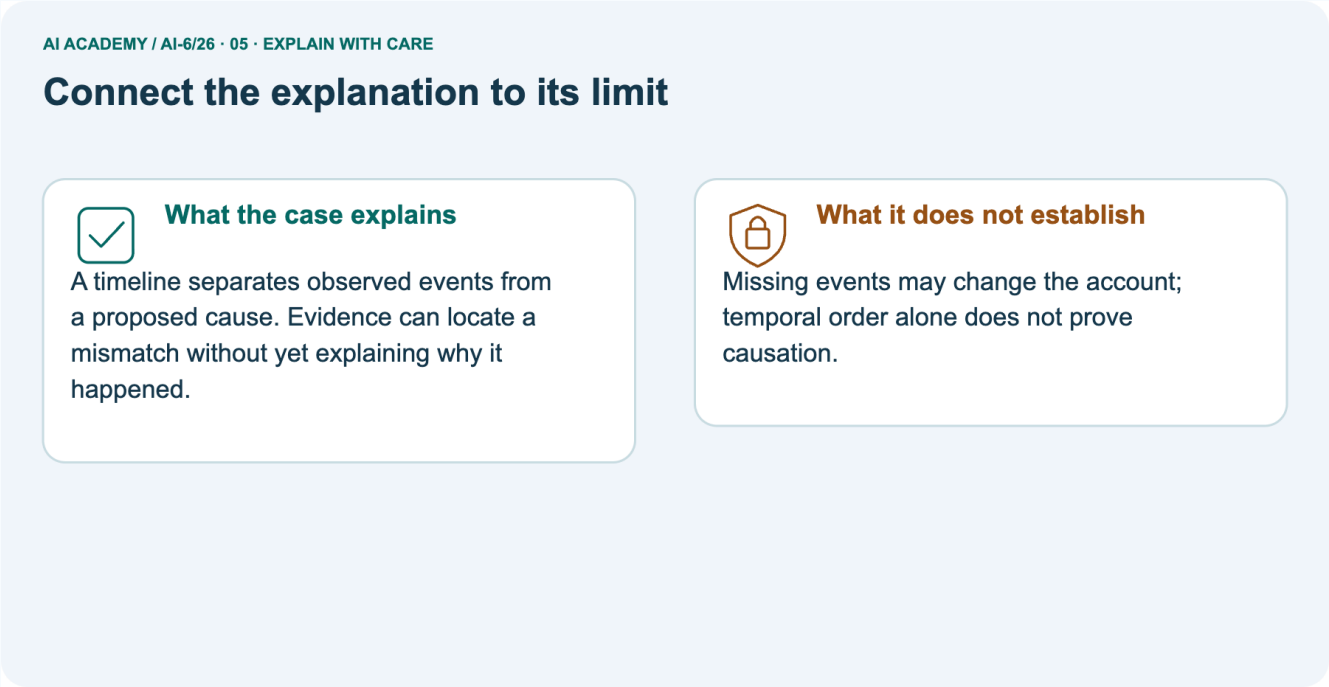
Original case: Reproduce version selection with controlled fixtures before naming the exact bug. Changed condition: Trace instead shows version 6 retrieved and room C drafted.

Expected result: Investigate drafting or validation before blaming caching. Reason: The mismatch now appears after retrieval.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit



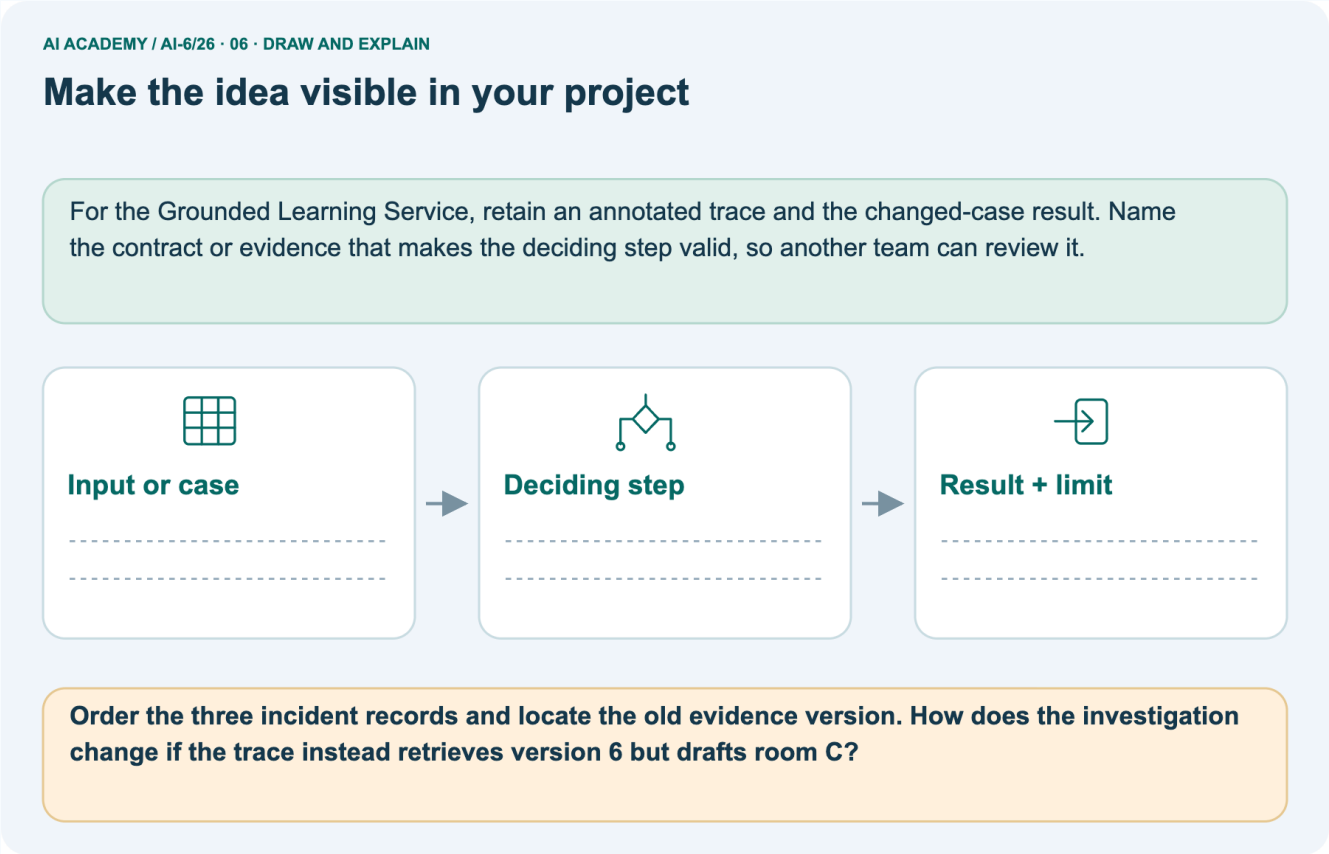

Follow the labelled picture with the matching explanation. Open the vector diagram

A timeline separates observed events from a proposed cause. **Evidence** can locate a mismatch without yet explaining why it happened. Limit: Missing events may change the account; temporal order alone does not prove causation.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Order the three incident records and locate the old **evidence** version.

How does the investigation change if the trace instead retrieves version 6 but drafts room C?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

## ▼ Earlier wording, alternatives and picture transcripts

**01. Incident investigation: the case**

**02. Follow the deciding steps**

**03. Incident investigation — evidence trace**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.